

untangle::async

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Chapter 1

async

c++ mechanism to run queued callables in an asynchronous fashion

The interface consists of one header file that exposes a generic execution class. It is a wrapper of a list of callables that are invoked on other thread. The callables are pumped into the list by invoking an action from the caller thread.

The implementation is using the **actuator** callable, and therefore it exemplifies how a generic callable may improve the code structure, by easily creating interfaces inside a class instead of using external ones.

There are provided two execution modes: one-off and continuous.

requirements

C++23. The interface reports warnings with `std::println`, so it needs a standard library that provides `<print>` — GCC 14 or newer; older `libc++` and `MSVC` releases may not have it yet.

one-off execution



example: one-off execution of a plain function

```
#include <async.hpp>

void f()
{
    std::cout << "f thread " << std::this_thread::get_id() << std::endl;
}

int main()
{
    std::cout << "main thread " << std::this_thread::get_id() << std::endl;

    // declare an execution specialized for an action type; it must be held by a shared pointer,
    // because an async binding keeps only weak ownership of it
    // the name is optional, and identifies the execution in the warnings it reports
    using void_execution = untangle::async::execution<std::function<void(void)>>;
    auto execution = void_execution::create_instance("oneoff");
```

```

std::function<void(void)> action;
// create async binding between the action and f; the execution is passed as a shared pointer,
// which the binding holds weakly - that is what lets the action outlive it safely
void_execution::bind_action_and_function(action, f, execution);
// whenever the action is invoked, a new callable wrapping f will be added to the execution's action list
action();
// run the execution
execution->run();

// add the execution object to the running poll
untangle::async::execution_poll::get().add(*execution);

// wait the polled executions to finish
while(untangle::async::execution_poll::get().is_running())
{
    std::this_thread::sleep_for(std::chrono::milliseconds(1));
}

return 0;
}

```

example: one-off execution of a class method

```

#include <async.hpp>

class A
{
public:
    void f(int x)
    {
        std::cout << "A::f " << x << " thread " << std::this_thread::get_id() << std::endl;
    }
    std::function<void(int)> action;
};

int main()
{
    std::cout << "main thread " << std::this_thread::get_id() << std::endl;

    auto a = std::make_shared<A>();

    // declare an execution specialized for an action type; it must be held by a shared pointer
    using int_execution = untangle::async::execution<std::function<void(int)>>;
    auto execution = int_execution::create_instance("oneoff_method");
    // note both the bound object and the execution must be shared pointers
    int_execution::bind_action_and_method(a->action, a, &A::f, execution);
    // whenever the action is invoked, a new callable wrapping f will be added to the execution's action list
    a->action(10);
    // run the execution
    execution->run();

    // add the execution object to the running poll
    untangle::async::execution_poll::get().add(*execution);

    // wait the polled executions to finish
    while(untangle::async::execution_poll::get().is_running())
    {
        std::this_thread::sleep_for(std::chrono::milliseconds(1));
    }

    return 0;
}

```

continuous execution



example: continuous execution of a plain function

```

#include <async.hpp>

void f()
{
    std::cout << "f thread " << std::this_thread::get_id() << std::endl;
}

```

```

}

int main()
{
    std::cout << "main thread " << std::this_thread::get_id() << std::endl;

    // declare an execution specialized for an action type; it must be held by a shared pointer
    using void_execution = untangle::async::execution<std::function<void(void)>>;
    auto execution = void_execution::create_instance("continuous");
    std::function<void(void)> action;
    // create async binding between the action and f
    void_execution::bind_action_and_function(action, f, execution);

    // run the execution
    execution->start();
    // add f to the execution's queue (three times)
    action();action();action();
    // stop the execution
    execution->stop();

    // add the execution object to the running poll
    untangle::async::execution_poll::get().add(*execution);

    // wait the polled executions to finish
    while(untangle::async::execution_poll::get().is_running())
    {
        std::this_thread::sleep_for(std::chrono::milliseconds(1));
    }

    return 0;
}

```

queueing and lifetime

Queueing is thread safe. An action may be invoked from any thread; the execution's list is guarded, and the worker is woken rather than polled for.

stop() ends the execution's working life. The worker runs whatever was queued before the call and then leaves. An action invoked after stop() is refused rather than queued, because it would never run, and the refusal is reported:

```
warning: execution 'continuous' is stopped, action not added
```

An action bound to an object that has since been destroyed is dropped the same way, rather than ending the process:

```
warning: execution 'continuous' dropped an invalid action: bind: invalid object
```

The worker is detached and cannot be joined. Waiting for an execution therefore means waiting on its running state, which is what execution_poll reports and what the examples above do. The destructor waits on the same state, so an execution that goes out of scope while its worker is still running blocks until the worker is finished rather than leaving it reading freed memory.

The poll holds a pointer to each execution added to it, so an execution withdraws itself in its destructor. execution_poll::remove() is available for withdrawing one earlier; calling it is not required.

building the tests

```

cmake -S test -B test/build
cmake --build test/build
ctest --test-dir test/build

```

test/CMakeLists.txt fetches googletest at configure time, so the first configure needs network access. The actuator submodule has to be present: git submodule update --init.

sanitizers

ASYNC_SANITIZE builds the tests under a sanitizer. It is off by default, because a sanitized build is several times slower and ThreadSanitizer does not ship for every toolchain.

```
cmake -S test -B test/build-asan -DASYNC_SANITIZE=address
cmake --build test/build-asan
ctest --test-dir test/build-asan
```

Accepted values are address, thread, undefined, or empty. Anything else is refused at configure time rather than passed through to the compiler.

Use a separate build directory per sanitizer. address and thread instrument the same accesses in incompatible ways and cannot be combined, so one build directory is one sanitizer.

The two answer different questions. ThreadSanitizer names both sides of a data race, which is how the queue's races were diagnosed. AddressSanitizer reports a use-after-free, which is how the dangling attachment was. Neither sees everything: a defect reached through a destroyed `std::mutex` throws `std::system_error` from inside `libsystem` before any instrumented load runs, so both sanitizers stay silent. Reach for a direct instrumented read when that happens.

Chapter 2

Topic Index

2.1 Topics

Here is a list of all topics with brief descriptions:

namespace untangle: functions [13](#)

Chapter 3

Namespace Index

3.1 Namespace List

Here is a list of all documented namespaces with brief descriptions:

untangle	Interface to untangle::async::execution class	15
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Chapter 4

Class Index

4.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

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Chapter 5

File Index

5.1 File List

Here is a list of all documented files with brief descriptions:

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Chapter 6

Topic Documentation

6.1 namespace untangle: functions

Chapter 7

Namespace Documentation

7.1 untangle Namespace Reference

Interface to [untangle::async::execution](#) class.

7.1.1 Detailed Description

Interface to [untangle::async::execution](#) class.

Chapter 8

Class Documentation

8.1 `untangle::async::attachment` Struct Reference

One execution's place in the attachment graph: who attached it, if anyone.

```
#include <async.hpp>
```

Public Attributes

- `std::weak_ptr< attachment >` `attacher`
The link of the execution that attached this one; empty when nothing has.

8.1.1 Detailed Description

One execution's place in the attachment graph: who attached it, if anyone.

At namespace scope, so it is one type rather than one per specialisation: an execution may be attached by an execution of any specialisation, and the chain is walked without knowing any link's action type. It dies with its execution, so it doubles as a liveness token - a link that still locks belongs to an execution that is still there.

The documentation for this struct was generated from the following file:

- `async.hpp`

8.2 `untangle::async::execution< actionT >` Class Template Reference

Asynchronous template execution class.

```
#include <async.hpp>
```

Public Member Functions

- `execution ()`
Constructs an execution with the default name. Call `create_instance()` instead.
- `execution (std::string exec_name)`
Constructs a named execution. Use `create_instance()` for one that is to be bound.
- `~execution ()`
Destroys the execution, once its worker has left.
- `bool is_running () const`
Is this execution still working?
- `bool is_busy () const`
Is this execution working through its action list?
- `void run ()`
Runs what is queued on a worker of its own, once, and keeps the results.
- `void start ()`
Starts a continuous worker that runs whatever is queued until `stop()`.
- `void stop ()`
Stops the worker, after the actions already queued have run.
- `template<typename... Args>`
`bool add_action (actionT action, Args... args)`
Queues an action, bound to args, and says whether it was taken.
- `template<typename otherT>`
`void attach (const std::shared_ptr< otherT > &other)`
Attaches another execution, so that this one triggers it.
- `template<typename otherT>`
`bool detach (otherT &other)`
Detaches an execution previously attached by `attach()`.
- `auto results () const`
The return values of the actions run by `run()`, in the order they ran.

Static Public Member Functions

- `template<typename... Args>`
`static std::shared_ptr< execution > create_instance (Args &&... args)`
Creates an execution owned by a `std::shared_ptr`, which is what binding to it requires.
- `template<typename classT, typename T>`
`static void bind_action_and_method (actionT &action, const std::shared_ptr< classT > &obj, T classT::*method, const std::shared_ptr< execution > &async_exec)`
Binds asynchronously an external action to a class function member.
- `template<typename T>`
`static void bind_action_and_function (actionT &action, const T &Fn, const std::shared_ptr< execution > &async_exec)`
Binds asynchronously an external action to a plain function.

Public Attributes

- `std::function< void(void)> action_execute`
Connection point that runs this execution's queued actions. Bound to `execute_actions()`.
- `std::function< void(void)> action_stop`
Connection point that stops this execution. Bound to `stop()`.
- `std::function< bool(void)> action_is_running`
Connection point that reports whether this execution is running. Bound to `is_running()`.
- `std::function< void(void)> on_finished`
Called on the worker's thread once the action list has drained. Assigned by the caller.

Static Public Attributes

- `static constexpr auto default_name = "default_name"`
Name of an execution built unnamed.

8.2.1 Detailed Description

```
template<typename actionT>
class untangle::async::execution< actionT >
```

Asynchronous template execution class.

An async execution provides a mechanism to queue actions and execute them sequentially on a separate thread. It may also attach another [execution](#) object and trigger its actions. This way actions of different types may be executed on the same thread.

The mechanism relies on an "asynchronous binding" created by [bind_action_and_method\(\)](#) or [bind_action_and_function\(\)](#).

Template Parameters

<code>actionT</code>	It represents the type of the action. It is specified as <code>std::function<...></code> and should match the signature of the bound function or class method.
----------------------	--

8.2.2 Constructor & Destructor Documentation

8.2.2.1 `execution()` [1/2]

```
template<typename actionT>
untangle::async::execution< actionT >::execution () [inline]
```

Constructs an execution with the default name. Call [create_instance\(\)](#) instead.

The name identifies this execution in the warnings it reports, and is otherwise unused.

8.2.2.2 `execution()` [2/2]

```
template<typename actionT>
untangle::async::execution< actionT >::execution (
    std::string exec_name) [inline], [explicit]
```

Constructs a named execution. Use [create_instance\(\)](#) for one that is to be bound.

Parameters

exec_name	- A name for this execution.
-----------	------------------------------

8.2.2.3 ~execution()

```
template<typename actionT>
untangle::async::execution< actionT >::~~execution () [inline]
```

Destroys the execution, once its worker has left.

The worker is detached and cannot be joined, so this waits on the "running" state instead. It detaches this execution from whatever attached it first, and returns at once for one that was never started.

Attention

Safe only because running_ is cleared as the very last thing the worker does: this object may be freed the moment it reads false, so nothing may follow it in execute() or loop().

Remarks

No rule of five, deliberately. An execution is declared once and used in place; the members already delete the copy operations several times over.

8.2.3 Member Function Documentation

8.2.3.1 create_instance()

```
template<typename actionT>
template<typename... Args>
std::shared_ptr< execution > untangle::async::execution< actionT >::create_instance (
    Args &&... args) [inline], [static]
```

Creates an execution owned by a std::shared_ptr, which is what binding to it requires.

An execution built any other way is still usable - [add_action\(\)](#) takes actions directly - but cannot be bound, and the attempt does not compile.

Parameters

args	- Constructor arguments: a name, or nothing for the default name.
------	---

Returns

- A std::shared_ptr owning the new execution.

8.2.3.2 bind_action_and_method()

```
template<typename actionT>
template<typename classT, typename T>
void untangle::async::execution< actionT >::bind_action_and_method (
    actionT & action,
    const std::shared_ptr< classT > & obj,
    T classT::* method,
    const std::shared_ptr< execution< actionT > > & async_exec) [inline], [static]
```

Binds asynchronously an external action to a class function member.

It creates an action as a binding to a class method (by `untangle::bind()`), and assigns to action a callable that passes it to `add_action()`. The execution is held weakly, so the action may outlive it.

Attention

Invoking action after the execution has been destroyed throws `untangle::invalid_action`. The worker catches it and drops the action with a warning.

Parameters

action	[in,out] - An action of type <code>std::function<...></code> .
obj	- A <code>std::shared_ptr</code> that wraps the bound class object.
method	- Pointer to function member. It is specified as <code>&<class type>::<function member></code> .
async_exec	- A <code>std::shared_ptr</code> owning the execution the action is queued on. Requiring it here is what keeps an execution that no <code>shared_ptr</code> owns from being bound at all.

8.2.3.3 bind_action_and_function()

```
template<typename actionT>
template<typename T>
void untangle::async::execution< actionT >::bind_action_and_function (
    actionT & action,
    const T & Fn,
    const std::shared_ptr< execution< actionT > > & async_exec) [inline], [static]
```

Binds asynchronously an external action to a plain function.

It wraps `Fn` in an action and assigns to action a callable that passes it to `add_action()`. The execution is held weakly, as in `bind_action_and_method()`.

Parameters

action	[in,out] - An action of type <code>std::function<...></code> .
Fn	- A plain function.
async_exec	- A <code>std::shared_ptr</code> owning the execution the action is queued on.

8.2.3.4 `is_running()`

```
template<typename actionT>
bool untangle::async::execution< actionT >::is_running () const [inline]
```

Is this execution still working?

False from the moment the worker starts reporting itself finished, so a caller polling it learns a run is over without waiting for the object to become safe to destroy.

Remarks

[on_finished](#) is raised from inside the drain and reads this as true - the worker is standing in the callback with the rest of its path still to go.

Returns

true - the execution is still working.

8.2.3.5 `is_busy()`

```
template<typename actionT>
bool untangle::async::execution< actionT >::is_busy () const [inline]
```

Is this execution working through its action list?

True from the moment an action is added until the last one has returned. Unlike [is_running\(\)](#), which is about the worker thread, this answers "has it room for more work"

- a worker started by [start\(\)](#) is running for its whole life whether busy or idle.

Remarks

Answers for this execution's own actions, not for those it triggers through [attach\(\)](#).

A false answer is durable only for the caller that is itself adding the actions; a true one may stop being true the moment after.

Returns

true - actions are queued, or one is running.

8.2.3.6 `run()`

```
template<typename actionT>
void untangle::async::execution< actionT >::run () [inline]
```

Runs what is queued on a worker of its own, once, and keeps the results.

The worker is detached: wait for it with [is_running\(\)](#) or [execution_poll](#). Actions queued while it is draining are run by it too. [results\(\)](#) is filled by this path only.

8.2.3.7 start()

```
template<typename actionT>
void untangle::async::execution< actionT >::start () [inline]
```

Starts a continuous worker that runs whatever is queued until `stop()`.

Unlike `run()`, the worker stays for the life of the execution, so `is_running()` is true whether or not there is work; ask `is_busy()` instead. Results are not collected.

8.2.3.8 stop()

```
template<typename actionT>
void untangle::async::execution< actionT >::stop () [inline]
```

Stops the worker, after the actions already queued have run.

Remarks

An action passed to `add_action()` after this call is refused, not queued: `stop()` is the end of this execution's working life, and an action accepted then would never run.

8.2.3.9 add_action()

```
template<typename actionT>
template<typename... Args>
bool untangle::async::execution< actionT >::add_action (
    actionT action,
    Args... args) [inline]
```

Queues an action, bound to args, and says whether it was taken.

It runs later, on whichever thread drives this execution - its own worker, or an attacher's. Queueing before `run()` or `start()` is normal.

Attention

A stopped execution refuses and drops the action, silently for a caller that ignores the answer. Nothing is thrown: `untangle::invalid_action` means a dead binding, while a refused action is sound and merely too late.

Remarks

A caller arriving through `bind_action_and_method()` or `bind_action_and_function()` never sees the answer: those lambdas return `actionT::result_type`, which has no room for it.

Template Parameters

Args	- The argument types the action is bound to.
------	--

Parameters

action	- The action to queue.
args	- The arguments to bind to action.

Returns

- true - queued, and this execution's worker will run it.
- false - the execution is stopped and the action was dropped.

8.2.3.10 attach()

```
template<typename actionT>
template<typename otherT>
void untangle::async::execution< actionT >::attach (
    const std::shared_ptr< otherT > & other) [inline]
```

Attaches another execution, so that this one triggers it.

The attached execution's pending actions are run by this one's worker, and stopping this one stops that one too. Actions of different types can be run on one thread this way.

The attachment is a pointer into other; what makes that safe is the record left on the other side, which [~execution\(\)](#) and [detach\(\)](#) use to take it back out.

Parameters

other	- A std::shared_ptr owning the execution to trigger; requiring one keeps an unowned execution from being attached at all. Attaching the same execution twice adds it twice, and one detach() still removes it completely.
-------	---

Exceptions

invalid_attachment	- if other is attached already. An execution has at most one attacher, which keeps the graph a forest and the cycle check a walk rather than a search.
invalid_attachment	- if the attachment would close a cycle, itself included. Driving a cycle recurses until the stack is gone, so it is refused where the caller can still be told.

8.2.3.11 detach()

```
template<typename actionT>
template<typename otherT>
bool untangle::async::execution< actionT >::detach (
    otherT & other) [inline]
```

Detaches an execution previously attached by [attach\(\)](#).

Both paths are unwired together, the triggering one and the stopping one.

Parameters

other	- The execution to stop triggering. Detaching one that was never attached is not an error: the caller asked for a state that already holds.
-------	---

Returns

- true - other was attached and is no longer.
- false - other was not attached to this execution.

8.2.3.12 results()

```
template<typename actionT>
auto untangle::async::execution< actionT >::results () const [inline]
```

The return values of the actions run by [run\(\)](#), in the order they ran.

Read it from [on_finished](#), which fires once the queue has drained, so the results are complete by the time the callback sees them. Returned by value: the worker is appending to it.

Remarks

Only [run\(\)](#) fills this. A continuous worker has no point at which a run is over, so [start\(\)](#) collects nothing.

Returns

The results of the most recent run, or empty if none has produced any.

8.2.4 Member Data Documentation

8.2.4.1 action_execute

```
template<typename actionT>
std::function<void(void)> untangle::async::execution< actionT >::action_execute
```

Connection point that runs this execution's queued actions. Bound to [execute_actions\(\)](#).

Attention

Wired by the constructor. Assigning to it unwires whatever is connected, silently.

8.2.4.2 `action_stop`

```
template<typename actionT>
std::function<void(void)> untangle::async::execution< actionT >::action_stop
```

Connection point that stops this execution. Bound to [stop\(\)](#).

[attach\(\)](#) wires it together with [action_execute](#), and [detach\(\)](#) unwires both.

Attention

Wired by the constructor. Assigning to it unwires whatever is connected, silently.

8.2.4.3 `action_is_running`

```
template<typename actionT>
std::function<bool(void)> untangle::async::execution< actionT >::action_is_running
```

Connection point that reports whether this execution is running. Bound to [is_running\(\)](#).

[execution_poll](#) holds its address, and [~execution\(\)](#) takes it back out.

Attention

Wired by the constructor. Assigning to it unwires whatever is connected, silently.

8.2.4.4 `on_finished`

```
template<typename actionT>
std::function<void(void)> untangle::async::execution< actionT >::on_finished
```

Called on the worker's thread once the action list has drained. Assigned by the caller.

[results\(\)](#) is complete by the time it runs, and [is_running\(\)](#) reads true there.

The documentation for this class was generated from the following file:

- `async.hpp`

8.3 `untangle::async::execution_poll` Class Reference

Single-tone poll that reports whether any [execution](#) added to it is still running.

```
#include <async.hpp>
```

Public Member Functions

- `template<typename asyncexecT>`
`void add (asyncexecT &async_exec)`
 Adds an [execution](#) object to the poll.
- `template<typename asyncexecT>`
`void remove (asyncexecT &async_exec)`
 Removes an [execution](#) object from the poll.
- `auto is_running ()`
 Checks if the execution objects are running.

Static Public Member Functions

- `static execution_poll & get ()`
 Gets the single-tone instance.

8.3.1 Detailed Description

Single-tone poll that reports whether any [execution](#) added to it is still running.

Remarks

An execution runs in a detached thread and cannot be joined, so checking the "running" state is the only way to wait for one. This checks any number of them at once.

8.3.2 Member Function Documentation

8.3.2.1 add()

```
template<typename asyncexecT>
void untangle::async::execution_poll::add (
    asyncexecT & async_exec) [inline]
```

Adds an [execution](#) object to the poll.

Parameters

<code>async_exec</code>	An execution object.
-------------------------	--------------------------------------

8.3.2.2 remove()

```
template<typename asyncexecT>
void untangle::async::execution_poll::remove (
    asyncexecT & async_exec) [inline]
```

Removes an [execution](#) object from the poll.

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Attention

[add\(\)](#) stores a pointer into the object, so one destroyed while still added would leave the poll calling

Parameters

async_exec	An execution object.
------------	--------------------------------------

8.3.2.3 is_running()

```
auto untangle::async::execution_poll::is_running () [inline]
```

Checks if the execution objects are running.

Returns

true - if at least one [execution](#) object in this poll is running.

8.3.2.4 get()

```
execution_poll & untangle::async::execution_poll::get () [inline], [static]
```

Gets the single-tone instance.

Returns

[execution_poll](#) An [execution_poll](#) instance.

The documentation for this class was generated from the following file:

- [async.hpp](#)

8.4 untangle::async::invalid_attachment Struct Reference

Raised by [execution::attach\(\)](#) when the attachment asked for is refused.

```
#include <async.hpp>
```

Public Member Functions

- [invalid_attachment](#) (std::string text)
Construct a new invalid attachment object.
- const char * [what](#) () const noexcept override
The message text describing the reason of this exception.

8.4.1 Detailed Description

Raised by [execution::attach\(\)](#) when the attachment asked for is refused.

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Remarks

Not an untangle::invalid_action: that one reports a binding whose target has died and is swallowed by the kernel, while this one reports a binding whose target has died and is not swallowed

Parameters

text	- A message text, describing the reason of this exception.
------	--

8.4.3 Member Function Documentation

8.4.3.1 what()

```
const char * untangle::async::invalid_attachment::what () const    [inline], [override], [noexcept]
```

The message text describing the reason of this exception.

Returns

const char* - The message text.

The documentation for this struct was generated from the following file:

- async.hpp

Chapter 9

File Documentation

9.1 async.hpp

```
00001 // Copyright (c) 2018 Nicolae Popescu. MIT License.
00002
00006 #pragma once
00007
00008 #include <actuator/actuator.hpp>
00009 #include <algorithm>
00010 #include <atomic>
00011 #include <chrono>
00012 #include <condition_variable>
00013 #include <exception>
00014 #include <functional>
00015 #include <list>
00016 #include <memory>
00017 #include <mutex>
00018 #include <print>
00019 #include <string>
00020 #include <thread>
00021 #include <type_traits>
00022 #include <utility>
00023 #include <vector>
00024
00025 namespace untangle {
00026 namespace async {
00030
00037 struct invalid_attachment : std::exception {
00043     explicit invalid_attachment(std::string text) : message_(std::move(text)) {}
00044
00050     const char* what() const noexcept override { return message_.c_str(); }
00051
00052 private:
00053     std::string message_;
00054 };
00055
00064 struct attachment {
00066     std::weak_ptr<attachment> attacher;
00067 };
00068
00075 class execution_poll {
00076 public:
00082     template <typename asyncexecT>
00083     void add(asyncexecT& async_exec) {
00084         std::lock_guard<std::mutex> lock(actuator_mutex_);
00085
00086         if (!actuator_is_running_.is_connected()) {
00087             actuator_is_running_ = untangle::connect(async_exec.action_is_running);
00088         } else {
00089             actuator_is_running_.add(&async_exec.action_is_running);
00090         }
00091     }
00092
00101     template <typename asyncexecT>
00102     void remove(asyncexecT& async_exec) {
00103         std::lock_guard<std::mutex> lock(actuator_mutex_);
00104
00105         actuator_is_running_.remove(&async_exec.action_is_running);
00106     }
00107 }
```

```

00113 auto is_running() {
00114     std::lock_guard<std::mutex> lock(actuator_mutex_);
00115     actuator_is_running_();
00116
00117     auto result = false;
00118     for (const auto& ret : actuator_is_running_.results) {
00119         result |= ret;
00120     }
00121     return result;
00122 }
00123
00129 static execution_poll& get() {
00130     static execution_poll instance;
00131     return instance;
00132 }
00133
00134 private:
00135 execution_poll() = default;
00136 ~execution_poll() = default;
00137 actuator<std::function<bool(void)>> actuator_is_running_;
00138 mutable std::mutex actuator_mutex_;
00139 };
00140
00154 template <typename actionT>
00155 class execution {
00156     // An execution may attach one of a different specialisation, which is a different class;
00157     // attach() and detach() reach into its attachment_lifetime_.
00158     template <typename otherActionT>
00159     friend class execution;
00160
00161 public:
00171     template <typename... Args>
00172     static std::shared_ptr<execution> create_instance(Args&&... args) {
00173         return std::make_shared<execution>(std::forward<Args>(args)...);
00174     }
00175
00181     execution() : execution(std::string(default_name)) {}
00182
00188     explicit execution(std::string exec_name) : name(std::move(exec_name)) {
00189         other_this_ = this;
00190         action_execute = untangle::bind(other_this_, &execution<actionT>::execute_actions);
00191         action_stop = untangle::bind(other_this_, &execution<actionT>::stop);
00192         action_is_running = untangle::bind(other_this_, &execution<actionT>::is_running);
00193     }
00194
00209 ~execution() {
00210     // Out of the attacher first, before the worker is even asked to stop: from here on nothing
00211     // may reach this object, and the actions about to be destroyed are the ones it holds.
00212     if (!attachment_lifetime_ -> attacher.expired()) {
00213         attacher_execute_ -> remove(&action_execute);
00214         attacher_stop_ -> remove(&action_stop);
00215     }
00216
00217     {
00218         std::lock_guard<std::mutex> lock(action_mutex_);
00219         started_ = false;
00220         stopped_ = true;
00221     }
00222
00223     action_cv_.notify_all();
00224
00225     while (running_.load()) { // time of check
00226         std::this_thread::sleep_for(std::chrono::milliseconds(1));
00227     }
00228
00229     // time of use: this object is freed once the destructor returns, so the check above is only
00230     // safe because running_ is the last thing the worker touches
00231     execution_poll::get().remove(*this);
00232 }
00233
00251 template <typename classT, typename T>
00252 static void bind_action_and_method(actionT& action, const std::shared_ptr<classT>& obj,
00253                                     T classT::* method,
00254                                     const std::shared_ptr<execution>& async_exec) {
00255     actionT async_action = untangle::bind(obj, method);
00256
00257     action = [wp = std::weak_ptr<execution>(async_exec),
00258             async_action](auto... args) -> actionT::result_type {
00259         // lock() also keeps the execution alive for the duration of the call
00260         const auto exec = wp.lock();
00261         if (!exec) {
00262             throw invalid_action("bind: invalid execution");
00263         }
00264         exec -> add_action(async_action, args...);
00265         return typename actionT::result_type();
00266     };
00267 }

```

```

00268
00279 template <typename T>
00280 static void bind_action_and_function(actionT& action, const T& Fn,
00281                                     const std::shared_ptr<execution>& async_exec) {
00282     actionT async_action = Fn;
00283
00284     action = [wp = std::weak_ptr<execution>(async_exec),
00285             async_action](auto... args) -> actionT::result_type {
00286         const auto exec = wp.lock();
00287         if (!exec) {
00288             throw invalid_action("bind: invalid execution");
00289         }
00290         exec->add_action(async_action, args...);
00291         return typename actionT::result_type();
00292     };
00293 }
00294
00306 bool is_running() const { return running_.load() && !finishing_.load(); }
00307
00323 bool is_busy() const {
00324     std::lock_guard<std::mutex> lock(action_mutex_);
00325     return !action_list_.empty() || executing_action_.load();
00326 }
00327
00334 void run() {
00335     running_ = true;
00336     finishing_ = false;
00337     collecting_results_ = true;
00338     this_thread_ = std::thread(&execution::execute, this);
00339     this_thread_.detach();
00340 }
00341
00348 void start() {
00349     running_ = true;
00350     finishing_ = false;
00351     collecting_results_ = false;
00352     {
00353         std::lock_guard<std::mutex> lock(action_mutex_);
00354         started_ = true;
00355         stopped_ = false;
00356     }
00357
00358     this_thread_ = std::thread(&execution::loop, this);
00359     this_thread_.detach();
00360 }
00361
00368 void stop() {
00369     {
00370         std::lock_guard<std::mutex> lock(action_mutex_);
00371         started_ = false;
00372         stopped_ = true;
00373     }
00374
00375     // The worker drains what is already queued before it leaves loop(), so stop() does not have to
00376     // spin on the list; waking it is enough.
00377     action_cv_.notify_all();
00378
00379     if (actuator_stop_.is_connected()) {
00380         actuator_stop_();
00381     }
00382 }
00383
00405 template <typename... Args>
00406 bool add_action(actionT action, Args... args) {
00407     {
00408         std::lock_guard<std::mutex> lock(action_mutex_);
00409
00410         // Once stopped, the worker is on its way out and would never reach this action; dropping it
00411         // here is what keeps it from sitting in the list looking as though it were pending.
00412         if (stopped_) {
00413             std::println(stderr, "warning: execution '{}' is stopped, action not added", name);
00414             return false;
00415         }
00416
00417         action_list_.push_back(std::bind(action, args...));
00418     }
00419
00420     action_cv_.notify_one();
00421     return true;
00422 }
00423
00443 template <typename otherT>
00444 void attach(const std::shared_ptr<otherT>& other) {
00445     if (static_cast<const void*>(other.get()) == static_cast<const void*>(this)) {
00446         throw invalid_attachment("attach: an execution cannot be attached to itself");
00447     }
00448 }

```

```

00449     if (!other->attachment_lifetime->attacher.expired()) {
00450         throw invalid_attachment("attach: that execution is attached already");
00451     }
00452
00453     // At most one attacher each makes the graph a forest, so a cycle closes exactly when other is
00454     // somewhere up this chain. Walking it is the whole check.
00455     for (auto link = attachment_lifetime->attacher.lock(); link; link = link->attacher.lock()) {
00456         if (link == other->attachment_lifetime) {
00457             throw invalid_attachment("attach: the attachment would close a cycle");
00458         }
00459     }
00460
00461     if (!actuator_execute_.is_connected()) {
00462         actuator_execute_ = untangle::connect(other->action_execute);
00463     } else {
00464         actuator_execute_.add(&other->action_execute);
00465     }
00466
00467     if (!actuator_stop_.is_connected()) {
00468         actuator_stop_ = untangle::connect(other->action_stop);
00469     } else {
00470         actuator_stop_.add(&other->action_stop);
00471     }
00472
00473     // What ~execution() needs to take itself back out of these actuators: the link proves this
00474     // attacher is still there, and the two pointers are only ever followed while it does.
00475     other->attachment_lifetime->attacher = attachment_lifetime_;
00476     other->attacher_execute_ = &actuator_execute_;
00477     other->attacher_stop_ = &actuator_stop_;
00478 }
00479
00491 template <typename otherT>
00492 bool detach(otherT& other) {
00493     // The actuator's own action list is the record of what is attached; nothing else has to keep
00494     // one. An action is stored as a pointer, so the attachment is found by identity.
00495     const auto& actions = actuator_execute_.actions;
00496     if (std::find(actions.begin(), actions.end(), &other.action_execute) == actions.end()) {
00497         return false;
00498     }
00499
00500     actuator_execute_.remove(&other.action_execute);
00501     actuator_stop_.remove(&other.action_stop);
00502
00503     other.attachment_lifetime->attacher.reset();
00504     other.attacher_execute_ = nullptr;
00505     other.attacher_stop_ = nullptr;
00506
00507     return true;
00508 }
00509
00521 auto results() const {
00522     std::lock_guard<std::mutex> lock(results_mutex_);
00523     return results_;
00524 }
00525
00526 static constexpr auto default_name = "default_name";
00527
00533 std::function<void(void)> action_execute;
00534
00542 std::function<void(void)> action_stop;
00543
00552 std::function<bool(void)> action_is_running;
00553
00559 std::function<void(void)> on_finished;
00560 std::string name = default_name;
00561
00562 private:
00563     using queued_action_t = std::function<typename actionT::result_type(void)>;
00564
00571     void execute_action(queued_action_t& action) {
00572         if constexpr (std::is_void_v<typename actionT::result_type>) {
00573             action();
00574         } else {
00575             auto value = action();
00576
00577             if (collecting_results_) {
00578                 std::lock_guard<std::mutex> lock(results_mutex_);
00579                 results_.push_back(std::move(value));
00580             }
00581         }
00582     }
00583
00593     void execute_actions() {
00594         actions_run_ = 0;
00595
00596         // The action is taken off the list under the lock and invoked with the lock released: an action
00597         // is caller code that may run for a while, and may itself call add_action().

```

```

00598     for (;;) {
00599         queued_action_t action;
00600
00601         {
00602             std::lock_guard<std::mutex> lock(action_mutex_);
00603             if (action_list_.empty()) {
00604                 break;
00605             }
00606
00607             action = std::move(action_list_.front());
00608             action_list_.pop_front();
00609
00610             // Under the lock that emptied the list, so is_busy() sees the pop and this together or
00611             // neither.
00612             executing_action_ = true;
00613         }
00614
00615         // Nothing may leave this loop: an exception escaping a detached thread function calls
00616         // std::terminate.
00617         try {
00618             execute_action(action);
00619         } catch (const invalid_action& ia) {
00620             std::println(stderr, "warning: execution '{}' dropped an invalid action: {}", name,
00621                 ia.what());
00622         } catch (const std::exception& e) {
00623             std::println(stderr, "warning: execution '{}' dropped an action that threw: {}", name,
00624                 e.what());
00625         } catch (...) {
00626             std::println(stderr, "warning: execution '{}' dropped an action that threw an unknown type",
00627                 name);
00628         }
00629
00630         executing_action_ = false;
00631
00632         // Counted whether it ran cleanly, reported a dead binding, or threw: it came off the queue,
00633         // and the queue is what the notification is about.
00634         ++actions_run_;
00635
00636         // The notification is raised without the lock: on_finished may call add_action(), which
00637         // takes it.
00638         bool drained = false;
00639         {
00640             std::lock_guard<std::mutex> lock(action_mutex_);
00641             drained = action_list_.empty();
00642         }
00643
00644         if (drained) {
00645             notify_finished();
00646         }
00647     }
00648
00649     // Last, and part of the pass: an attachment is only ever reached through action_execute, which
00650     // is this function, so whoever drives this execution drives the ones attached to it.
00651     if (actuator_execute_.is_connected()) {
00652         actuator_execute_();
00653     }
00654 }
00655
00660 void notify_finished() {
00661     if (actions_run_ == 0 || !on_finished) {
00662         return;
00663     }
00664
00665     {
00666         // An action may queue another, so a pass that drained can leave more behind it. That is the
00667         // next batch, not the end of this one.
00668         std::lock_guard<std::mutex> lock(action_mutex_);
00669         if (!action_list_.empty()) {
00670             return;
00671         }
00672     }
00673
00674     on_finished();
00675 }
00676
00677 void execute() {
00678     // The results belong to this run, not to this run and the last one.
00679     {
00680         std::lock_guard<std::mutex> lock(results_mutex_);
00681         results_.clear();
00682     }
00683
00684     execute_actions();
00685
00686     // After the drain and the callback it raised: there is no more work to be told about.
00687     finishing_ = true;
00688 }

```

```

00689     std::println("execution '{}{}' finishing thread", name);
00690
00691     // Must stay last: ~execution() may free this object the moment it reads false.
00692     running_ = false;
00693 }
00694
00695 void loop() {
00696     for (;;) {
00697         {
00698             std::unique_lock<std::mutex> lock(action_mutex_);
00699
00700             // Bounded: an attached execution has its own list and cannot notify this condition
00701             // variable.
00702             action_cv_.wait_for(lock, std::chrono::milliseconds(10),
00703                 [this] { return !action_list_.empty() || !started_; });
00704
00705             if (!started_ && action_list_.empty()) {
00706                 break;
00707             }
00708         }
00709         execute_actions();
00710     }
00711
00712     // A last pass for the attachments; this execution's own list is already empty here.
00713     execute_actions();
00714
00715     // After the last pass, because that pass still runs work.
00716     finishing_ = true;
00717
00718     std::println("execution '{}{}' thread finished", name);
00719
00720     // Must stay last: ~execution() may free this object the moment it reads false.
00721     running_ = false;
00722 }
00723
00724 std::thread this_thread_;
00725
00726 std::atomic_bool started_ = {false};
00727 std::atomic_bool running_ = {false};
00728
00729 // Set by stop(), and the difference between "not started yet" and "finished for good": actions
00730 // may be queued before run()/start(), but not after stop().
00731 std::atomic_bool stopped_ = {false};
00732
00733 // action_list_, started_ and stopped_ are written by every thread that calls add_action() or
00734 // stop() and read by the worker; nothing touches them outside this mutex.
00735 mutable std::mutex action_mutex_;
00736 std::condition_variable action_cv_;
00737
00738 std::list<std::function<typename actionT::result_type(void)>> action_list_;
00739
00740 using void_actuator = actuator<std::function<void(void)>>;
00741
00742 void_actuator actuator_execute_;
00743 void_actuator actuator_stop_;
00744
00745 std::shared_ptr<attachment> attachment_lifetime_ = std::make_shared<attachment>();
00746
00747 void_actuator* attacher_execute_ = nullptr;
00748 void_actuator* attacher_stop_ = nullptr;
00749
00750 // std::vector cannot hold void type; use an arbitrary type e.g. int
00751 using resultT = std::conditional<std::is_void<typename actionT::result_type>::value, int,
00752     typename actionT::result_type>;
00753
00754 std::vector<typename resultT::type> results_;
00755
00756 mutable std::mutex results_mutex_;
00757
00758 std::atomic_bool collecting_results_ = {false};
00759
00760 std::atomic_bool finishing_ = {false};
00761
00762 std::atomic_bool executing_action_ = {false};
00763
00764 std::atomic_size_t actions_run_ = {0};
00765
00766 execution* other_this_;
00767 };
00768 } // namespace async
00769 } // namespace untangle

```

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